SMART INDIA HACKATHON 2025

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TITLE PAGE

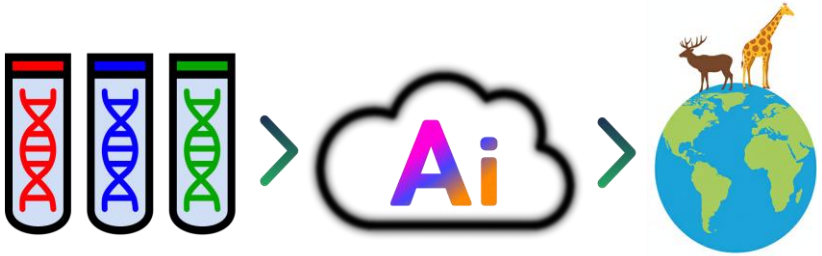
- Problem Statement ID: **25042**
- Problem Statement Title:

  Identifying Taxonomy and Assessing Biodiversity from eDNA Datasets
- Theme: **Miscellaneous**
- PS Category: **Software**
- Team ID: **56745**
- Team Name: **GENOMIKS**

IDENTIFYING TAXONOMY AND ASSESSING BIODIVERSITYFROM eDNA DATASETS

Our Idea:

A cloud-based AI-powered pipeline that takes raw eDNA data and outputs Species Taxonomy + Biodiversity Health Report.

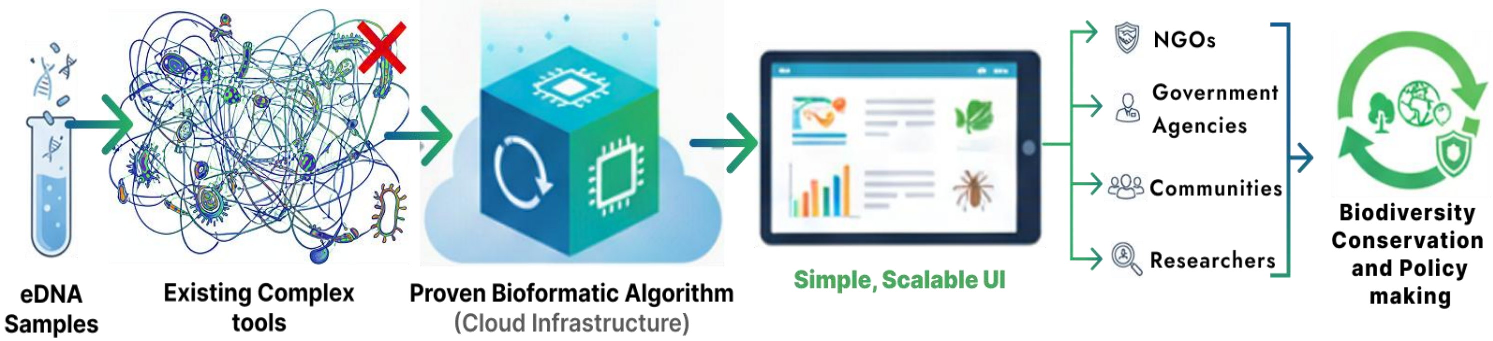


How its Innovative?

The solution's key uniqueness is its ability to produce a user-friendly biodiversity health report in an easily digestible format.

Our 5 Step Solution

1. **Data Input** : Upload raw eDNA sequencing files.
2. **Preprocessing & Noise filtering** : Remove low-quality reads, sequencing errors, and contamination.
3. **Taxonomy Identification** : Integrate a **deep learning model** trained on reference genomes. Comparison against reference databases.
4. **Biodiversity Assessment** : Calculate indices and species. **Detect invasive/endangered species.**
5. **Visualization Dashboard** : **Interactive UI, Species list, Graphs** and options to **export reports as PDF or CSV.**



TECHNICAL APPROACH

Methodology & Required Technologies

1. Collect & Process data → Collection + Splitting → Processing: Spark, N8n | Docker, Hadoop → Database: MongoDB
2. Taxonomy Classification → Ai Model → Training + Validation → Tools: Python, TensorFlow, Bio python

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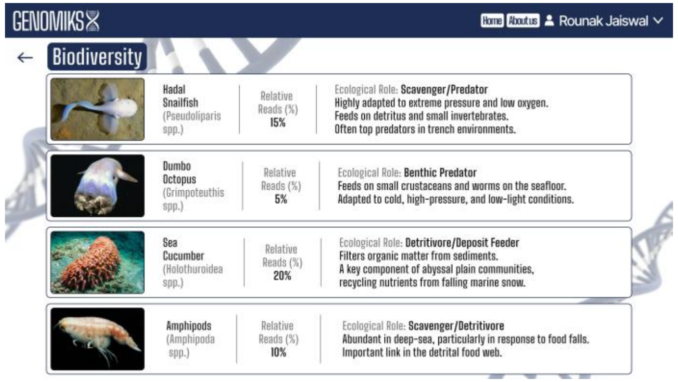
3. Biodiversity Computation → Data Aggregation & Calculation → Tools: Python, TensorFlow, Bio python
4. Dashboard Design → Tools: Figma, HTML/CSS | JavaScript, Firebase
5. Report Generation → Generative AI
6. Cloud Deployment → Containerization → AWS / GCP / Azure

Our Prototype

Zoom to see the how the outputs will be shown



[Click here](#) for Prototype

FEASIBILITY AND VIABILITY

FEASIBILITY ANALYSIS

- Existing tools are slow, complex, and require expertise. Our system offers a fast, scalable, and affordable alternative by leveraging cloud resources.

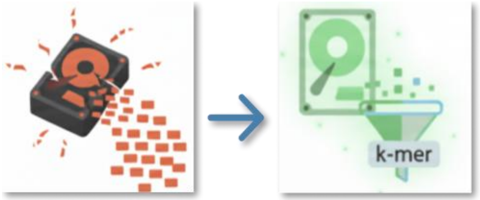
- The algorithm directly aids biodiversity conservation, making it useful for government agencies, NGOs, researchers, and local communities.

VIABILITY ANALYSIS

Potential Challenges and Solutions:

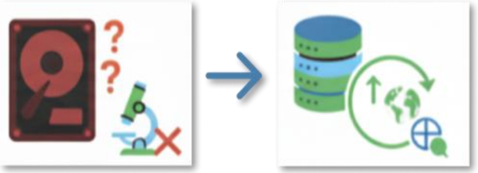
❑ **Large Data Size:** Processing raw sequencing files is computationally heavy.

✓ **Solution: Use of k-mer based classification for faster matching.**



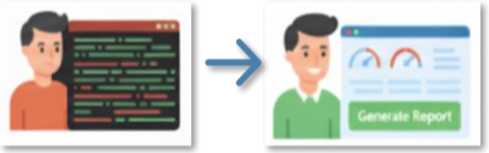
❑ **Accuracy of Taxonomy Identification:** Short DNA fragments propose a misclassification risk.

✓ **Solution: Hybrid approach and regular updates from global databases.**



❑ **Limited Bioinformatics Expertise Among End Users:** Biologists/NGOs cannot handle raw command-line tools.

✓ **Solution: Intuitive dashboard UI, Auto-generate Biodiversity Reports.**



GENOMIKS

IMPACTS AND BENEFITS

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IMPACTS

Environmental impact:

Enables real-time monitoring of ecosystems and Helps detect endangered, invasive, or extinct species faster.

Societal & Community impact:

Assists fishermen, farmers, and local communities by detecting ecosystem health early. Creates awareness among citizens about India's biodiversity wealth.

Economic impact:

Cost-effective solution for government agencies, NGOs, and industries monitoring biodiversity. Supports eco-tourism and natural resource management by ensuring healthy ecosystems.

BENEFITS

Non-invasive monitoring:

No need to capture or disturb organisms; biodiversity can be tracked from water/soil/air samples.

High accuracy:

Detects even hidden, rare, or microscopic species that traditional surveys miss.

Scalability:

Works for local rivers as well as large marine ecosystems using cloud infrastructure.

User-friendly platform:

Biologists, NGOs, and policymakers can use it without deep bioinformatics knowledge.

Supports National missions:

Contributes to India's biodiversity, climate, and sustainability goals.

RESEARCH AND REFERENCES

Existing Tools:
Qiime 2:- https://qiime2.org/
Blast:- https://blast.ncbi.nlm.nih.gov/Blast.cgi

Dataset Sources:
NCBI:- https://ftp.ncbi.nlm.nih.gov/blast/db/
BOLD:- https://boldsystems.org/data/data-packages/

Taxonomy Classification:
NCBI:- https://www.ncbi.nlm.nih.gov/taxonomy
ITIS:- https://www.itis.gov/